running a single virtual machine, you will spin up multiple machines in different data centers, also known as availability zones. Next, you will learn how to increase the resilience of your code. Afterward, you will create an infrastructure consisting of a queue (SQS), a load balancer (ALB), EC2 instances managed by Auto Scaling groups, and a database (DynamoDB).

16.1 Using redundant EC2 instances to increase availability

Here are just a few reasons your virtual machine might fail:

- If the host hardware fails, it can no longer host the virtual machine on top of it.
- If the network connection to/from the host is interrupted, the virtual machine will lose the ability to communicate over the network.
- If the host system is disconnected from the power supply, the virtual machine will fail as well.

Additionally, the software running inside your virtual machine may also cause a crash for the following reasons:

- If your application contains a memory leak, the EC2 instance will run out of memory and fail. It may take a day, a month, a year, or more, but eventually it will happen.
- If your application writes to disk and never deletes its data, the EC2 instance will run out of disk space sooner or later, causing your application to fail.